

Training Configuration and Summary:

Sequencing:

- timesteps: 32
- batch_size: 256

Model Hyperparameters:

- min_units: 16
- max_units: 64
- units_step: 16
- dropout_min: 0.2
- dropout_max: 0.5
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: lstm_hyperparameter_tuning

LSTM Training Report

Report Generated: 2025-04-09_09-30-27

Training Start: 2025-04-09 09:29:15

Training End: 2025-04-09 09:29:59

Training Duration: 44.84 seconds

Best Hyperparameters:

Number of Layers: 1

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 48 units

Performance Metrics:

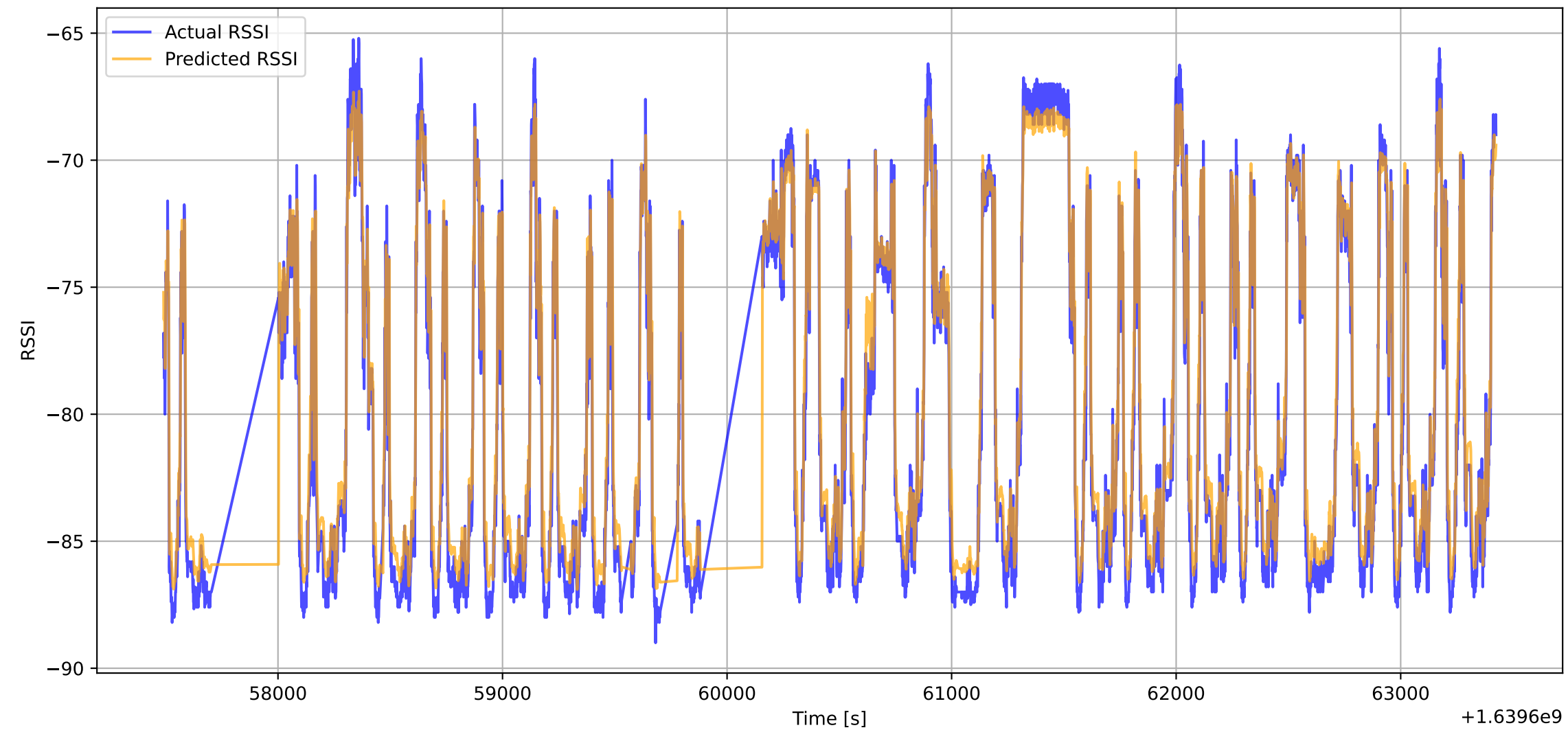
MAE: 1.1502

MSE: 2.2143

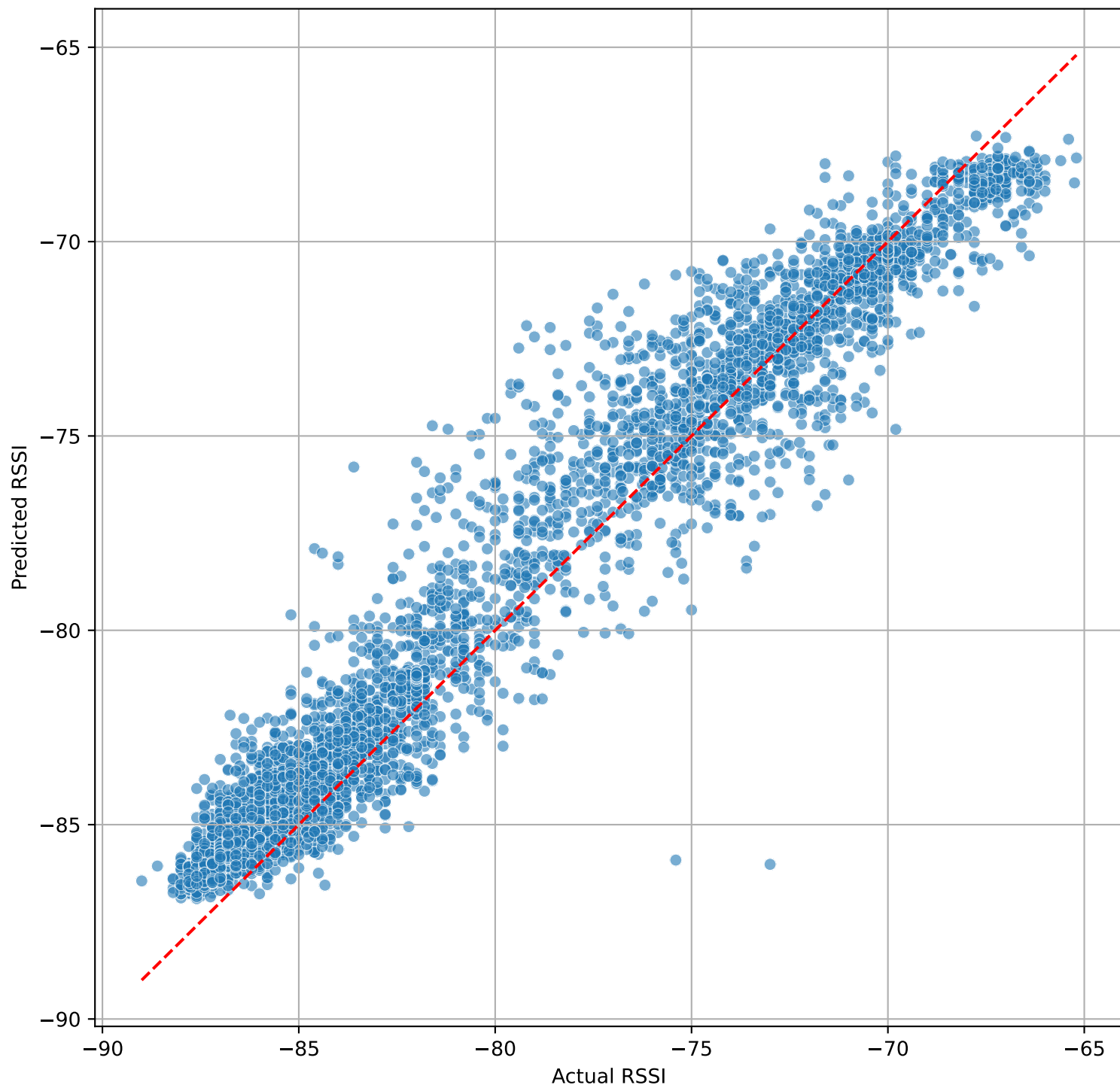
RMSE: 1.4880

R2 Score: 0.9497

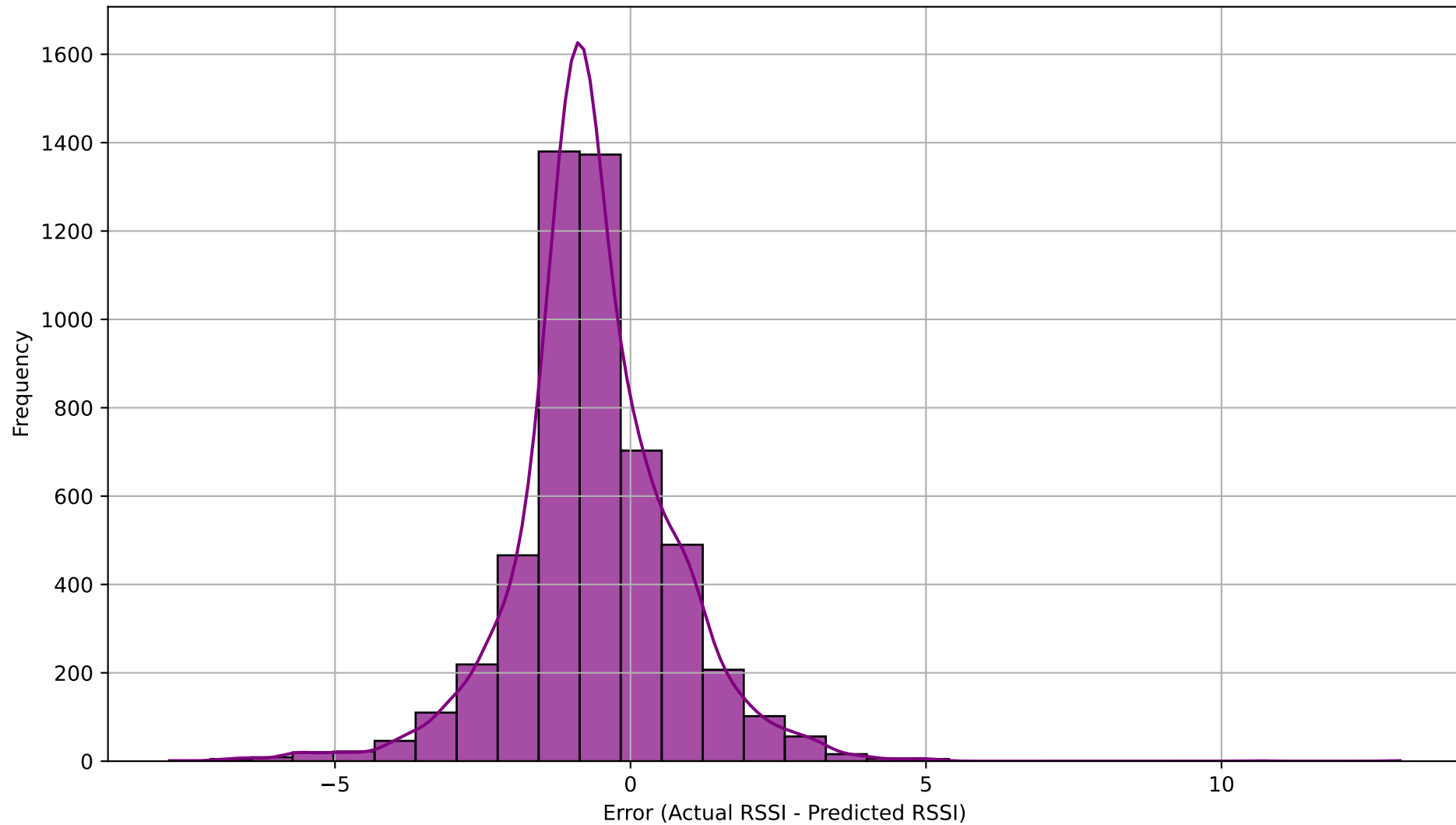
Actual vs Predicted RSSI Over Time



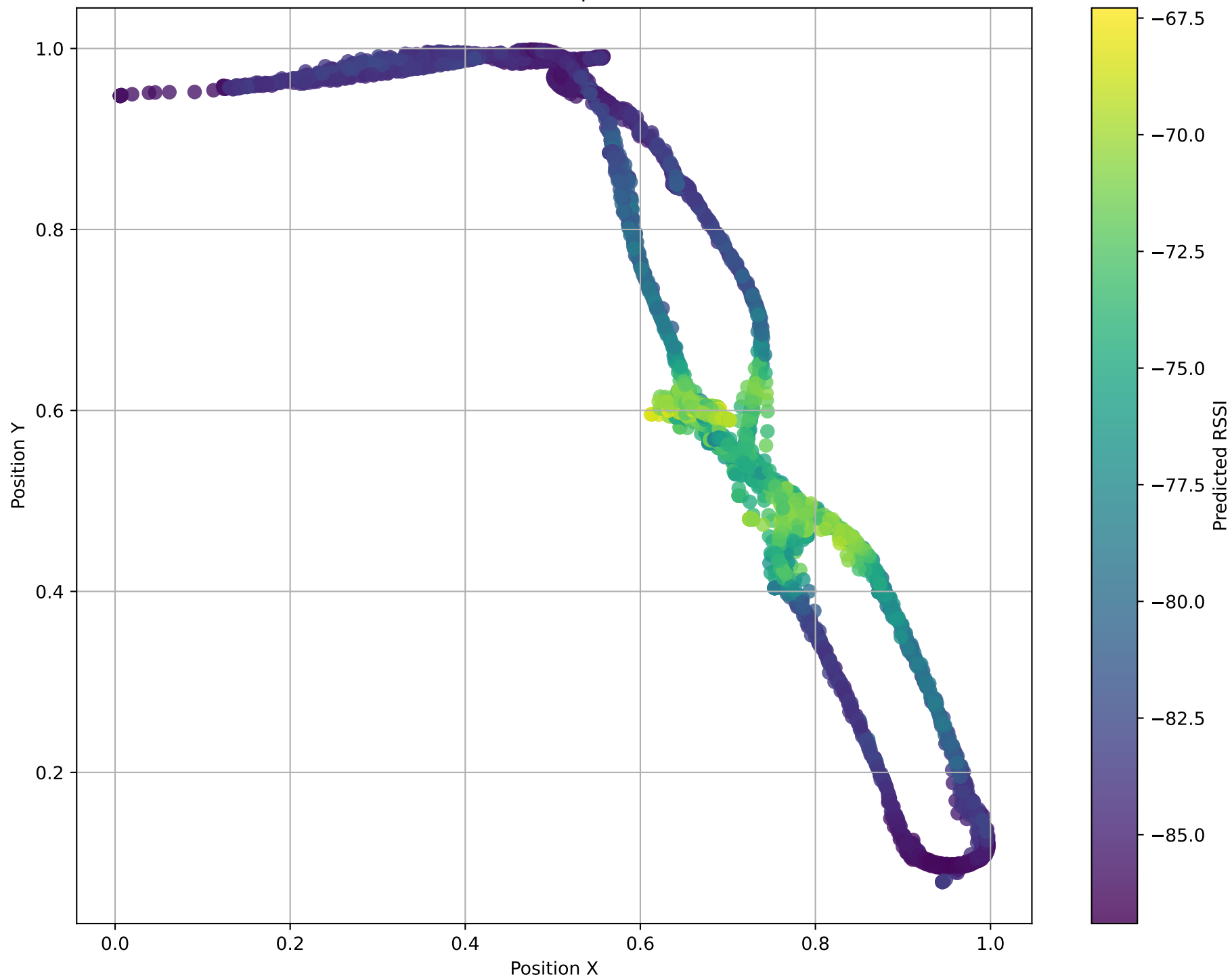
Actual vs Predicted RSSI



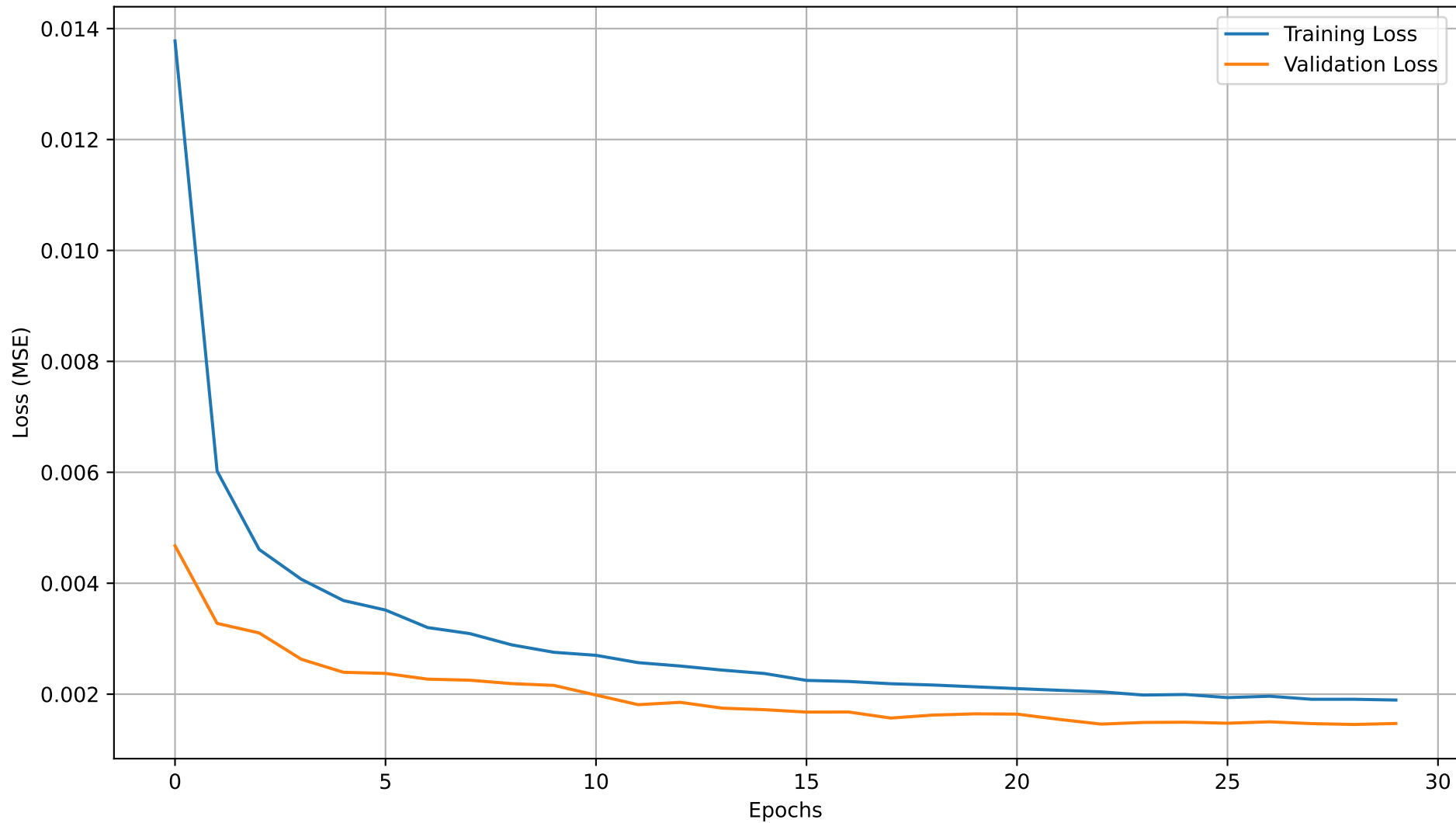
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

